

FarmConnect Project Development Report

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Repository root: 'farmconnect/'

1. Executive Summary

FarmConnect is a monorepo AgriTech marketplace platform that connects producers and buyers, with role-based workflows for buyers, producers, and admins. The implementation uses a web SPA ('apps/web'), a REST API ('apps/api'), and a separate AI microservice ('apps/ai-service'), with shared TypeScript contracts in workspace packages.

Primary capabilities implemented so far include:

- Authentication with access/refresh token flow.
- Role-aware dashboards and protected routes.
- Product catalog publication, listing, filtering, location-aware sorting, and favorites.
- Cart, orders, payment integration hooks, and payout entities.
- Reviews/ratings, reports/moderation inputs, notifications, analytics, and admin controls.
- Delivery tracking entities and APIs.
- AI recommendation/forecast/chatbot integration endpoints.
- Internationalization (Arabic/RTL and multilingual resources) and theme support.

2. Monorepo Technology Stack

- Workspace tooling: pnpm workspaces + Turborepo.
- Web: React 19, Vite, Tailwind, React Router, TanStack Query, Zustand.
- API: Express + TypeScript, Prisma ORM, PostgreSQL.
- AI Service: FastAPI + Python 3.12.
- Shared contracts: '@farmconnect/shared' types, enums, schemas.

3. Exact Folder Structure

3.1 Full snapshot

The exact raw full tree snapshot was generated directly from the filesystem with:

```
'tree /F /A farmconnect > farmconnect/docs/_full-tree.txt'
```

This preserves the literal folder/file structure at report time, including generated/cache folders.

Reference file:

- 'docs/_full-tree.txt'

3.2 Functional structure (human-oriented)

- 'apps/'
 - 'api/': Main backend service (REST, auth, business logic, Prisma).
 - 'web/': Frontend SPA for all roles.
 - 'ai-service/': AI microservice for forecasting/recommendations/chatbot.
- 'packages/'
 - 'shared/': Shared enums/types/Zod schemas/constants used by API+Web.
 - 'ui/': Reusable UI primitives/components.
 - 'config/': Shared ESLint/Prettier/TS base configs.
- 'docs/': Architecture, deployment runbooks, this report, and structure snapshots.
- 'infra/': Deployment/infrastructure assets (if present in current branch).
- 'scripts/': Utility/dev automation scripts.
- '.github/workflows/': CI/CD workflows for API/Web/AI/db-migrate.

4. Key File and Folder Purpose Map

4.1 Root-level files

- 'package.json': Workspace scripts and top-level dependencies.
- 'pnpm-workspace.yaml': Workspace package discovery.
- 'turbo.json': Turborepo pipeline config.
- '.env.example': Baseline environment variables.
- 'README.md': Setup, run, quality gates, deployment pointers.

4.2 API app ('apps/api')

- 'src/app.ts': Express app setup, middleware, CORS, versioned routing.
- 'src/config/routes.ts': Main API module routing table.
- 'src/modules/*': Domain modules using controller/service/repository/policy layers.
- 'prisma/schema.prisma': Full relational schema and enums.
- 'src/middleware/*': auth, rate limit, logging, locale, error/404 handling.

Registered API domains (from 'src/config/routes.ts'):

- 'health', 'auth', 'profile', 'products', 'cart', 'orders', 'payments', 'notifications', 'delivery', 'analytics', 'admin', 'ai', 'reviews', 'favorites', 'coupons', 'reports'.

4.3 Web app ('apps/web')

- 'src/app/router/index.tsx': Browser router composition.
- 'src/app/router/routes/*': Public/dashboard/admin route definitions and guards.
- 'src/features/*': Feature-first modules (auth, products, cart, orders, reports, admin, etc.).
- 'src/locales/*': Translation resources.
- 'src/app/providers/*': Auth/query/theme/i18n/notifications providers.

4.4 AI service ('apps/ai-service')

- FastAPI endpoints for AI-assisted marketplace capabilities:
 - Recommendations.
 - Demand/price forecasting.
 - Chatbot support flow.

4.5 Shared packages

- 'packages/shared': Cross-app contract source of truth (role enums, DTO types, schemas).
- 'packages/ui': Shared React UI layer.
- 'packages/config': Shared code style/tooling setup.

5. Implemented System Functionalities

5.1 Authentication and authorization

- JWT access token + rotating refresh token model.
- Refresh token persistence and session lifecycle controls.
- Role-based access enforcement (buyer/producer/admin).
- Protected routes in frontend and guarded API endpoints.

5.2 User and profile management

- Buyer and producer profile entities.
- Producer verification workflow.
- Address and geo fields to support location-aware flows.
- User status lifecycle (active/suspended/banned-related statuses in schema).

5.3 Product catalog and discovery

- Producer product CRUD and status management.
- Category hierarchy.
- Product images and logs/history.
- Public marketplace listing.
- Filters: query text, category, wilaya, min/max price, tags, availability, offers, favorite producers.
- Sorting: newest, price asc/desc, rating desc, distance asc.
- Optional geolocation-assisted nearest sorting.

5.4 Favorites and social trust

- Favorite products capability.
- Favorite producers capability.
- Buyer reviews/ratings model (for product and producer order context).

5.5 Cart, orders, and checkout

- Cart and cart-item management.
- Order creation and itemization.
- Coupon model support.
- Payment workflow modules and event tracking.
- Payout model for producer settlements.

5.6 Notifications, reports, moderation, admin

- In-app notification model/channel abstraction.
- Report submission and lifecycle (target/reason/status).
- Admin module for moderation and operational controls.
- Audit logging model and actions.

5.7 Delivery and logistics support

- Delivery status and method enums.
- Delivery tracking entity linked to orders.

5.8 Analytics and AI

- Daily analytics snapshot model.
- AI recommendation and forecast persistence.
- Dedicated AI service integration route/module.

5.9 UX and platform quality

- i18n with RTL support for Arabic UX.
- Theme support infrastructure.
- Security middleware: helmet, CORS strategy, rate limiting, request-id tracing.
- CI/CD workflows for build/test/deploy and DB migration jobs.

6. Database Schema (Prisma) Summary

Source: 'apps/api/prisma/schema.prisma'

6.1 Core identity and account entities

- 'User': central identity, role, status, and core account fields.
- 'RefreshToken': long-lived session token storage/rotation.
- 'ProducerProfile', 'BuyerProfile': role-specific profile extensions.
- 'ProducerVerificationRequest': compliance/verification pipeline.
- 'Address': user-linked location records.

6.2 Commerce entities

- 'Category': hierarchical product taxonomy.
- 'Product': producer listings with pricing/stock/geo/status.
- 'ProductImage': product media attachments.
- 'ProductLog': lifecycle/activity log per product.
- 'Cart', 'CartItem': pre-order basket model.
- 'Order', 'OrderItem': final transaction structure.
- 'Coupon': discount rules/codes.

6.3 Trust and engagement entities

- 'Review': rating/review records tied to buyer/order/product/producer context.
- 'FavoriteProduct', 'FavoriteProducer': user preference links.
- 'Report': abuse/reporting system with typed targets and statuses.

6.4 Payments and settlement entities

- 'Payment': payment intent/charge representation.
- 'PaymentEvent': append-only payment event timeline.
- 'Payout': producer payout settlement tracking.

6.5 Operations entities

- 'DeliveryTracking': delivery progress updates.
- 'Notification': user notification records.
- 'AnalyticsDaily': daily aggregated KPIs.
- 'AuditLog': actor/action audit history.

6.6 AI data entities

- 'AiRecommendation': persisted recommendation outputs.
- 'AiForecast': stored forecast results.

6.7 Schema enums (selected groups)

- Identity/access: 'Role', 'UserStatus', 'VerificationStatus'.
- Commerce lifecycle: 'ProductStatus', 'OrderStatus', 'ProductUnit', 'ProductLogType'.
- Financial lifecycle: 'PaymentStatus', 'PaymentMethod', 'PaymentEventType', 'PayoutStatus', 'CouponType'.
- Delivery/notifications: 'DeliveryStatus', 'DeliveryMethod', 'NotificationType', 'NotificationChannel'.
- Governance: 'AuditAction', 'ReportTargetType', 'ReportReason', 'ReportStatus'.

7. API and Web Routing Evidence

- API route mountpoint list is defined in 'apps/api/src/config/routes.ts'.
- App-level middleware and router registration are defined in 'apps/api/src/app.ts'.
- Web route composition is in 'apps/web/src/app/router/index.tsx', backed by route modules in 'apps/web/src/app/router/routes/'.

8. Architecture and Decisions Already Documented

- 'docs/architecture.md': layer responsibilities, auth model, phased roadmap.
- 'docs/architecture-decisions.md': ADRs for monorepo, shared contracts, AI microservice split, Prisma/PostgreSQL, idempotency, observability, and deployment baseline.

9. CI/CD and Operational Readiness

- CI workflow: '.github/workflows/ci.yml'.
- Service deployment workflows: 'api-cd.yml', 'web-cd.yml', 'ai-cd.yml'.
- DB migration workflow: 'db-migrate.yml'.
- Deployment/runbook docs: 'docs/deployment-baselines.md', 'docs/migration-deploy-runbook.md'.

10. Current Implementation Notes

- The codebase includes both application source and local runtime/generated artifacts ('node_modules', '.turbo', '.runlogs'), which are reflected in the full tree snapshot.
- This report is generated from repository files present in the current working copy and does not infer undocumented external infrastructure.

11. Recommended Appendices

- Appendix A: Full exact tree ('docs/_full-tree.txt').
- Appendix B: Complete Prisma schema source ('apps/api/prisma/schema.prisma').
- Appendix C: Complete API source file index ('apps/api/src/**').
- Appendix D: Complete Web source file index ('apps/web/src/**').

12. Conclusion

FarmConnect currently provides a substantial end-to-end marketplace foundation with role-based flows, catalog/discovery, ordering/payment entities, moderation/admin modules, and AI integration scaffolding. The architecture is modular, documented, and aligned with production-oriented practices (security middleware, CI/CD, migration workflow, auditability), making it

suitable for continued hardening and feature completion.